Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2024-25

Class:	SE	**Semester:**	III
Course Code:	CSL304	**Course Name:**	OBJECT-ORIENTED PROGRAMMING IN JAVA

Name of Student:	SHRUTI GAUCHANDRA
Roll No. :	15
Assignment No.:	02
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	10

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge Legibility	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : MS. NEHA RAUT

Signature :

Date :

Q1. Differentiate between Object-Oriented Programming and Procedure-Oriented Programming.

Ans:

	Object-Oriented Programming	Procedure-Oriented Programming
① Concept	It is based on the concept of objects which are instances of classes.	It is based on the concept of procedures which are also known as functions.
② Structure	Code is organised around objects and classes. It emphasises grouping related data & functions together.	Code is organised into functions and the program is sequence of function calls
③ Key principles/ characteristics	• Encapsulation • Abstraction • Inheritence • Polymorphism	• Modularization • Sequential Execution
④ Data handling	Focuses on data rather than functions, meaning data is tightly. integrated with functions that operate on it.	Focuses on functions rather than data. Data is often passed from one function to another leading to less emphasis on data encapsulation
⑤ Common languages which use them	• Java • C++ • Python • C#	• C • Pascal • Fortran

FOR EDUCATIONAL USE

Q2. Explain different feature of java & a role of JVM with a neat labelled diagram.

Ans: Java is a high level, class based, object-oriented programming language that is designed to have as few implentation dependencies as possible. It is general purpose programming language intended to let programmers write once, run anywhere; meaning that compiled Java code can run on all platforms that support Java without the need of recompilati-on.

Features of Java are as follows:

① Platform Independent:
Java is designed to be platform independent. The principle 'Write Once Run Anywhere' (WORA) means that Java code can run on any device that has Java Virtual Machine (JVM)

② Simple and easy to learn:
Java syntax is designed to be simple and easy to understand.

③ Robust and secure:
Java design priorities reliability and robusters. Includes features like exception handling and strong memory management, reducing the chances of system crashes. Security features in

Java absence of explicit pointers and secure runtime environment contribute to building secure applications.

④ Dynamic & Extensible:
Java's dynamic nature allows developers to adapt and modify the code during runtime. The extensibility contributes to creation of scalable and adaptable software.

⑤ Multithreading:
Java supports multithreading which allows multiple threads to run concurrently. This is useful for performing multiple operations at once improving performance.

⑥ Portable:
Java Bytecode is portable across different platforms making it easy to distribute and run multiple Java applications on various devices.

Role of Java Virtual Machine (JVM):
The JVM is essential for executing Java programs. The JVM also manages memory through garbage collection, enforce security with bytecode verificat-ion, optimizes performance through Just-In-Time (JIT) compilation

JVM diagram

Bytecode Interperetation
Just-In-Time Compilation
Memory Management
Security
Platform Independent

} Role of JVM

Diagram of Java Virtual Machine:

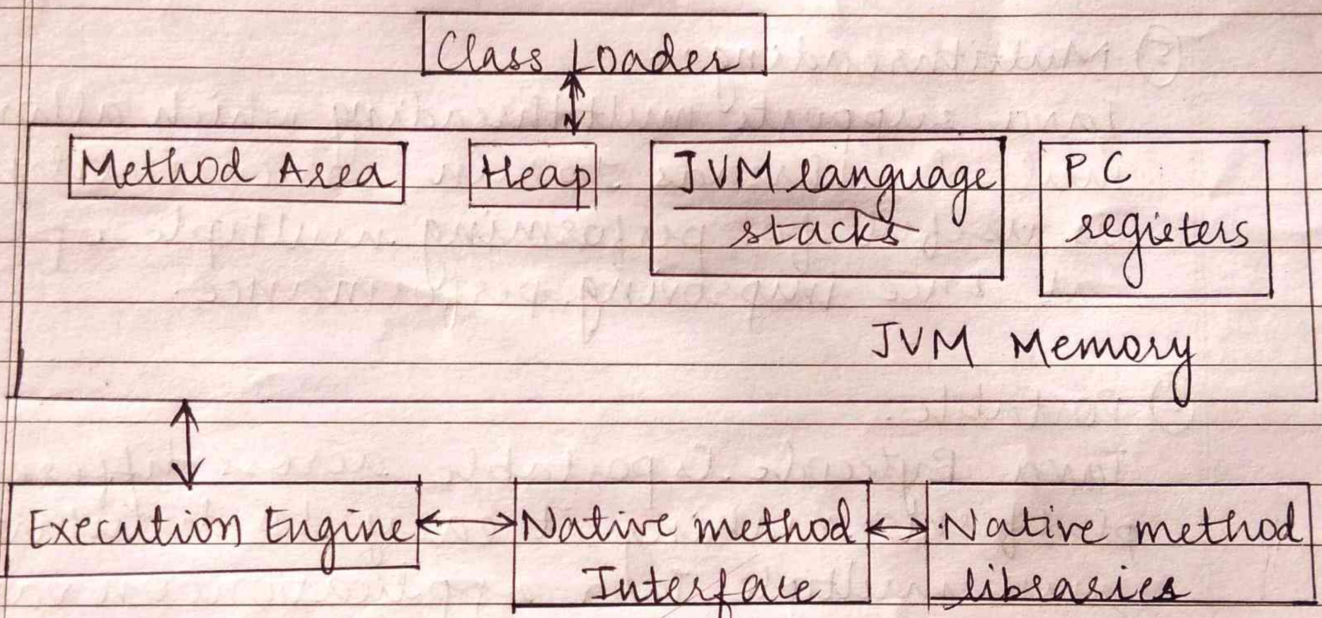
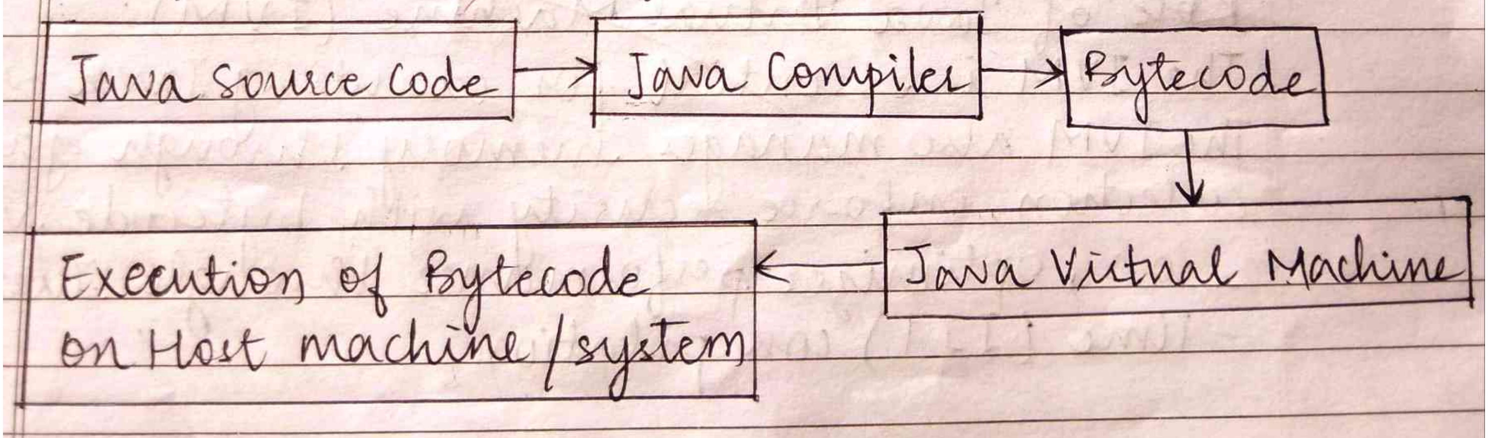


Diagram of Java Program Execution:



Q3. Some examples of data to be stored are listed below with the data type. Mention the data type that will be best suited for them.

(i) Age in years.
→ Integers (int) is the data type best suited respectively.

(ii) Rate of Interest
→ Floating-point number (float/double) are the data types best suited respectively.

(iii) Area of circle
→ Floating-point number (float/double) are the best suited data types respectively.

(iv) Runs made by a batsman.
→ Integer (int) is the best suited data type respectively.

(v) User input as 'true' or 'false'.
→ Boolean (boolean) is the best suited data type respectively.